

RAPHAEL SULZER



CONTACT

✉ raphaelsulzer@gmx.de
🏠 raphaelsulzer.de
📍 Nice, France

🌐 LinkedIn
🐙 GitHub
👤 Google Scholar
🔍 ResearchGate

SKILLS

Programming

Python	9 yrs
C++	8 yrs
Bash	5 yrs
HTML/CSS	4 yrs
C#	2 yrs
JavaScript	2 yrs

Languages

German	Native
English	Fluent
French	Conversational
Dutch	Beginner

Other

3D modeling	aws	Blender
CGAL	Cesium	CMake
computer vision	conda	
deep learning	Docker	GIS
geometry processing	GitHub	
LaTeX	LiDAR	machine learning
matplotlib	mesh	numpy
photogrammetry	point cloud	
pytorch	unix	vscode

EXPERIENCE

RESEARCH ENGINEER

📅 11/2023 - Present
📍 LUXCARTA, MOUANS-SARTOUX, FRANCE
Research and development of modern 2D and 3D mapping solutions.

POSTDOCTORAL RESEARCHER

📅 11/2022 - Present
📍 TITANE, INRIA, SOPHIA-ANTIPOLIS, FRANCE
Research in geometry processing, computer vision and deep learning.

PHD RESEARCHER

📅 12/2018 - 10/2022
📍 LASTIG, INSTITUT GÉOGRAPHIQUE NATIONAL, PARIS, FRANCE
📍 IMAGINE, ÉCOLE DES PONTS PARISTECH, MARNE-LA-VALLÉE, FRANCE
Research in geometry processing, computer vision and deep learning.

LECTURER

📅 12/2018 - 12/2019
📍 ÉCOLE NATIONALE DES SCIENCES GÉOGRAPHIQUES, PARIS, FRANCE
Design and implementation of e-learning courses in photogrammetry and GIS.

GIS DEVELOPER

📅 05/2018 - 07/2019
📍 GEO-COL GIS AND COLLABORATIVE PLANNING, AMSTERDAM, NETHERLANDS
Development of custom GIS applications for public and private customers.

GRADUATE STUDENT INTERN

📅 06/2017 - 02/2018
📍 ARUP, AMSTERDAM, NETHERLANDS
Development of an algorithm for building classification from remote sensing and cadastral data.

EDUCATION

PHD DEGREE, DEEP LEARNING AND GEOMETRY PROCESSING

📅 12/2018 - 10/2022
📍 GUSTAVE EIFFEL UNIVERSITY, MARNE-LA-VALLEÉE, FRANCE
PhD Thesis: Learning Surface Reconstruction from Point Clouds in the Wild ([link](#))

MASTER'S DEGREE, MSC GEOMATICS, CUM LAUDE

📅 08/2016 - 05/2018
📍 DELFT UNIVERSITY OF TECHNOLOGY, DELFT, NETHERLANDS
Master's Thesis: Shape Based Classification of Seismic Building Structural Types ([link](#))

ERASMUS EXCHANGE SEMESTER

📅 08/2015 - 06/2016
📍 DELFT UNIVERSITY OF TECHNOLOGY, DELFT, NETHERLANDS

MSC GEODESY AND GEOINFORMATICS

📅 05/2015 - 01/2016
📍 UNIVERSITY OF STUTTGART, STUTTGART, GERMANY

BACHELOR'S DEGREE, BSC GEODESY AND GEOINFORMATICS

📅 10/2011 - 05/2015
📍 UNIVERSITY OF STUTTGART, STUTTGART, GERMANY
Bachelor's Thesis: Photogrammetric Measurement of Snow Depth Using an UAV Platform ([link](#))

REFERENCES

Florent LAFARGE

 florent.lafarge@inria.fr

 TITANE, INRIA

 Postdoctoral advisor

Loïc LANDRIEU

 loic.landrieu@enpc.fr

 IMAGINE, ÉCOLE DES PONTS PARISTECH

 Doctoral advisor

PUBLICATIONS

Concise Plane Arrangements for Low-Poly Surface and Volume Modelling

 R. Sulzer, F. Lafarge

 2024  European Conference on Computer Vision

 [arXiv](#)

SimpliCity: Reconstructing Buildings with Simple Regularized 3D Models

 J.-P. Bauchet, R. Sulzer, F. Lafarge, Y. Tarabalka

 2024  CVPR Workshop on Urban Scene Modeling

 [arXiv](#)

A Survey and Benchmark of Automatic Surface Reconstruction from Point Clouds

 R. Sulzer, L. Landrieu, R. Marlet, B. Vallet

 2024  IEEE Transactions on Pattern Analysis and Machine Intelligence

 [arXiv](#)

Evaluating Surface Mesh Reconstruction Using Real Data

 Y. Marchand, L. Caraffa, R. Sulzer, E. Clédat, B. Vallet

 2023  Photogrammetric Engineering & Remote Sensing Journal, Volume 89, Number 10

 [link](#)

Deep Surface Reconstruction from Point Clouds with Visibility Information

 R. Sulzer, L. Landrieu, A. Boulch, R. Marlet, B. Vallet

 2022  26th International Conference on Pattern Recognition (ICPR), Montréal, Québec

 [arXiv](#)

Scalable Surface Reconstruction with Delaunay-Graph Neural Networks

 R. Sulzer, L. Landrieu, R. Marlet, B. Vallet

 2021  Computer Graphics Forum, Wiley, 2021, Eurographics Symposium on Geometry Processing 2021

 [arXiv](#)

Shape Based Classification of Seismic Building Structural Types

 R. Sulzer, P. Nourian, M. Palmieri, J. van Gemert

 2018  International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, 2018

 [link](#)

Track-id: Activity Determination based on Wi-Fi Monitoring

 van der Spek, S., Verbree, E., Quak, W., Groeneveld, I. J. D. G., Sulzer, R., Theocharous, E., Xu, Y.

 2016  Proceedings of the 13th International Conference on Location Based Services: LBS 2016

 [link](#)